Milestone1

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Using simulated data modelled on OECD unemployment statistics, this project plans to compare tertiary-educated foreign-born and native-born women across more and less developed regions.

Unemployment Rate among Highly-Educated Immigrant Women by the Country of Destination

Introduction

Questions

Immigrants with a university degree are often expected to secure stable positions in the labor market, yet a degree does not guarantee the same outcome across nations. Immigrants may face disadvantages depend on where they settle. For this research, I focus on highly educated female immigrants and ask: **Does the unemployment rate of tertiary-educated foreign-born women vary by the development level of the destination country? Specifically, is the unemployment rate lower if immigrant women migrate to more developed countries?**

Data

For this project, I plan to use **2023 OECD data on unemployment rates** (OECD 2023). The data comes from the OECD Data Explorer (DSD_EAG_LSO_EA@DF_LSO_NEAC_UNEMP_MIGR). Each row is a subgroup in one country, defined by variables such as sex, age group, education level and place of birth, together with its unemployment rate. The data contain other columns, which I do not list here.

My outcome variable will be the unemployment rate. I plan to keep tertiary-educated women, group countries into regions, and then group the regions into "more developed" and "less developed" regions. I will exclude Israel because it is difficult to classify as either a more or less developed region, and I also exclude the OECD average because it is not a country.

Expected Results

As more developed countries may have more established systems for recognizing foreign qualifications, I expect highly educated foreign-born women to experience lower unemployment in Northern America, Oceania, Western Europe and Northern Europe, which I group as "more developed" regions, than in Latin America, Southern Europe and Eastern Europe, which I group as "less developed" regions.

I will treat the hypotheses as supported if the mean unemployment rate is lower in more developed than in less developed regions by about 1 percentage point or more. If the differences are small, I will conclude that there is no clear pattern.

Limitations

Although the OECD data provides detailed data, it does not report immigrants' country of origin. Where immigrants come from may affect whether their educational credentials are recognized in the country of destination, and therefore their unemployment rate. As a result, differences in unemployment rates may partly reflect differences in the origins of their immigrant populations, rather than characteristics of the destination countries themselves.

```
library(tidyverse)
library(tinytex)
library(knitr) #for kable
```

PART I: Simulating Data

```
set.seed(916)

#Simulate the structure of the real data, based on the data of OECD
country_region <- tibble(
  country = c("Canada", "United States",
              "Mexico", "Costa Rica", "Colombia", "Chile",
              "Australia", "New Zealand",
              "Austria", "Belgium", "France", "Germany", "Luxembourg",
              "Netherlands", "Switzerland",
              "Estonia", "Ireland", "Latvia", "Lithuania", "Norway",
              "Sweden", "United Kingdom",
              "Greece", "Italy", "Portugal", "Slovenia", "Spain",
```

```
               "Bulgaria", "Czechia", "Hungary", "Poland", "Romania",
               "Slovak Republic"),
  region  = c(rep("Northern America", 2),
               rep("Latin America", 4),
               rep("Oceania", 2),
               rep("Western Europe", 7),
               rep("Northern Europe", 7),
               rep("Southern Europe", 5),
               rep("Eastern Europe", 6))
)

head(country_region)
```

```
# A tibble: 6 x 2
  country       region
  <chr>         <chr>
1 Canada        Northern America
2 United States Northern America
3 Mexico        Latin America
4 Costa Rica    Latin America
5 Colombia      Latin America
6 Chile         Latin America
```

```
#add characteristics
n_countries <- length(country_region$country)

simulated_ppl <- tibble(
  country = rep(country_region$country, each = 18),
  sex = rep(rep(c("Female", "Male"), each = 9), times = n_countries),
  education = rep(rep(c("Below upper secondary",
                        "Upper secondary / post-secondary non-tertiary",
                        "Tertiary"), each = 3),
                  times = 2 * n_countries),
  birth_place = rep(c("Foreign-born", "Foreign-born", "Native-born"),
                    times = 6 * n_countries),
)

head(simulated_ppl)
```

```
# A tibble: 6 x 4
  country sex    education                                    birth_place
```

```
  <chr>   <chr>  <chr>                                        <chr>
1 Canada  Female Below upper secondary                      Foreign-born
2 Canada  Female Below upper secondary                      Foreign-born
3 Canada  Female Below upper secondary                      Native-born
4 Canada  Female Upper secondary / post-secondary non-tertiary Foreign-born
5 Canada  Female Upper secondary / post-secondary non-tertiary Foreign-born
6 Canada  Female Upper secondary / post-secondary non-tertiary Native-born
```

```
#add region and development group
simulated_groups <-
  simulated_ppl |>
  left_join(country_region, by = "country") |>
  mutate(
    development_group = case_match(
      region,
      c("Northern America", "Oceania", "Western Europe", "Northern Europe") ~ "More developed",
      c("Latin America", "Southern Europe", "Eastern Europe") ~ "Less developed",
      .default = NA_character_
    )
  )

head(simulated_groups)
```

```
# A tibble: 6 x 6
  country sex    education                  birth_place region development_group
  <chr>   <chr>  <chr>                      <chr>       <chr>  <chr>
1 Canada  Female Below upper secondary      Foreign-bo~ North~ More developed
2 Canada  Female Below upper secondary      Foreign-bo~ North~ More developed
3 Canada  Female Below upper secondary      Native-born North~ More developed
4 Canada  Female Upper secondary / post-se~ Foreign-bo~ North~ More developed
5 Canada  Female Upper secondary / post-se~ Foreign-bo~ North~ More developed
6 Canada  Female Upper secondary / post-se~ Native-born North~ More developed
```

```
#add unemployment rate
simulated_rates <-
  simulated_groups |>
  mutate(
    unemployment_rate = sample(seq(1, 15, by = 0.1),
                               size = n(),
                               replace = TRUE)
  )
```

```
head(simulated_rates)
```

```
# A tibble: 6 x 7
  country sex   education birth_place region development_group unemployment_rate
  <chr>   <chr> <chr>     <chr>       <chr>  <chr>                           <dbl>
1 Canada  Fema~ Below up~ Foreign-bo~ North~ More developed                  8.8
2 Canada  Fema~ Below up~ Foreign-bo~ North~ More developed                  9.9
3 Canada  Fema~ Below up~ Native-born North~ More developed                 11
4 Canada  Fema~ Upper se~ Foreign-bo~ North~ More developed                  5.6
5 Canada  Fema~ Upper se~ Foreign-bo~ North~ More developed                  5.2
6 Canada  Fema~ Upper se~ Native-born North~ More developed                  6.7
```

```
simulated_rates |>
  summarise(min_rate =
            min(unemployment_rate),
            max_rate = max(unemployment_rate))
```

```
# A tibble: 1 x 2
  min_rate max_rate
     <dbl>    <dbl>
1        1     14.9
```

```
simulated_data <-
  simulated_rates |>
  select(region, development_group, sex, education, birth_place, unemployment_rate)
```

```
women_tertiary_simulated <-
  simulated_data |>
  filter(sex == "Female",
         education == "Tertiary") |>
  drop_na(development_group, unemployment_rate)

#only foreign-born women for the mean
mean_by_group <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(development_group) |>
  summarise(mean_rate = mean(unemployment_rate))

mean_by_group
```

```
# A tibble: 2 x 2
  development_group mean_rate
  <chr>                <dbl>
1 Less developed        7.33
2 More developed        8.33
```

Table

Table 1 shows 10 randomly selected rows of the simulated data, where each row is a subgroup in one country, defined by region and place of birth, together with its simulated unemployment rate.

```
simulated_data |>
  head(10) |>
  select(region, development_group, birth_place, unemployment_rate) |>
  kable(
    col.names = c("Region", "Group", "Place of birth", "Unemployment rate (%)"),
    caption = "Simulated data"
  )
```

Table 1: Simulated data

Region	Group	Place of birth	Unemployment rate (%)
Northern America	More developed	Foreign-born	8.8
Northern America	More developed	Foreign-born	9.9
Northern America	More developed	Native-born	11.0
Northern America	More developed	Foreign-born	5.6
Northern America	More developed	Foreign-born	5.2
Northern America	More developed	Native-born	6.7
Northern America	More developed	Foreign-born	9.5
Northern America	More developed	Foreign-born	3.0
Northern America	More developed	Native-born	9.2
Northern America	More developed	Foreign-born	3.1

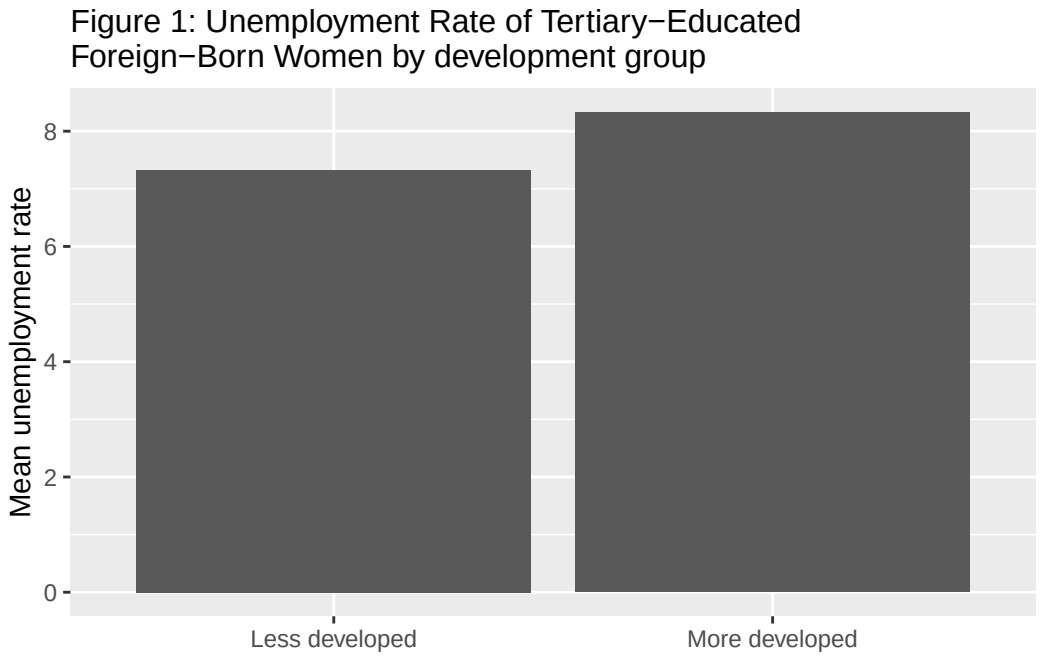
PART II: Graph

The following figures use the simulated data to show what the patterns would look like if the data followed each hypothesis.

Figure

Figure 1 compare the simulated mean unemployment rate of tertiary-educated foreign-born women in more and less developed regions.

```
#H1a
mean_by_group |>
  ggplot(aes(x = development_group, y = mean_rate)) +
  geom_col() +
  labs(x = NULL,
       y = "Mean unemployment rate",
       title = "Figure 1: Unemployment Rate of Tertiary-Educated\nForeign-Born Women by devel
  theme(plot.title = element_text(size = 12))
```



Reference

OECD. 2023. "DSD_EAG_LSO_EA@DF_LSO_NEAC_UNEMP_MIGR." OECD Data Explorer. https://data-explorer.oecd.org/vis?lc=en&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_EAG_LSO_EA%40DF_LSO_NEAC_UNEMP_MIGR&df[ag]=OECD.EDU.IMEP.NOBSERVATIONS&lo=1&pd=2023,2023&to[TIME_PERIOD]=true&lb=id&vw=ov.